

Why do lower-income areas experience worse road safety outcomes? Examining the role of the built environment in Orange County, Florida

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ABSTRACT

Lower-income areas experience an increased incidence of traffic crashes, injuries, and deaths than more affluent ones. Researchers tend to blame these outcomes on differences in car ownership, with lower-income households expected to use vulnerable active modes at higher rates, as well as being more likely to drive older vehicles with fewer safety features. While these explanations seem plausible enough, they likely fail to provide a complete picture of the problem. This study examines the role of the built environment in road safety outcomes for lower- and higher-income block groups in Orange County, Florida. It finds that the hazard posed by urban arterials is three times greater for lower-income environments than for more affluent communities. Sidewalk buffers, ordinarily regarded as a pedestrian amenity, were associated with crash increases in affluent areas but not lower-income ones. Areas with concentrations of black residents were found to be increased risk, even after accounting for differences in income. Considered as a whole, the risk factors for lower-income and high-income populations are not the same. This article examines the underlying reasons that lead to these outcomes and discusses the need to better account for the manner in which design may uniquely affect the safety of different demographic cohorts.

1. Introduction

Existing research consistently finds that areas with concentrations of lower-income populations experience an increased incidence of traffic-related crashes, injuries, and deaths (Abdalla et al., 1997; Baker et al. 2002; Chichester et al., 1998; Cottrill and Thakuria, 2010; Graham et al., 2005; Hippisley-Cox et al., 2002; Marshall and Ferenchak, 2017; Rifaat et al., 2010; Roberts and Powers, 1996; Valverde and Jovanis, 2006). Surprisingly, there has been little research into what is behind these trends. A common explanation suggests that lower-income residents are less likely to own cars. Or if lower-income residents do own a car, the car might be shared with others and only available intermittently (Blumenberg and Manville, 2004; King et al., 2022; Murakami and Young, 1997; Noland et al., 2013). Both circumstances are expected to lead to the increased use of more vulnerable modes such as walking and bicycling (Merlin et al., 2020). Another explanation is that lower-income populations are more likely to drive older vehicle that lack contemporary safety features, which may result in increased crash frequency or severity (Blumenberg and Haas, 2002; Cervero et al., 2002; County of Los Angeles, 2000; Murakami and Young, 1997; Ong and

Houston, 2002). Such arguments make intuitive sense, but they may not be telling the whole story.

A growing body of research suggests a significant association between road safety outcomes and characteristics of the built environment, which includes street design, sidewalks, and land use (Cai et al., 2017; Dumbaugh and Rae, 2009; Dumbaugh and Li, 2010; Dumbaugh and King, 2018; Marshall and Garrick, 2011; Schneider et al., 2021). This research has tended to focus on road safety outcomes generally, rather than examining whether specific environmental features may have different effects based on different demographic groups. The tendency to examine populations in aggregate can overlook the unique vulnerabilities that may place different groups at disproportionate risk of death and injury. To date, there has been little comparative examination of environmental risk factors in lower-income and higher-income communities. One study found that motorists in lower-income areas were more likely to yield to a pedestrian in a crosswalk than those in higher-income areas, though the authors note that street design may explain these differences, as the high-income street had more lanes and higher posted speeds than the lower-income comparison site (Coughenour et al., 2017). Nonetheless, most research examining the subject

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typically simply treats income as a control measure when examining other risk factors, such as roadway or community design, rather than a specific focus of concern.

But what if the differences in crash incidence are not simply a matter of degree, but also a matter of kind? Is it reasonable to assume that the nature of risk in lower-income and higher-income environments are roughly the same? Might behavioral differences in lower-income and higher-income environments result in certain environmental features posing disproportionate risk to different groups and, if so, what is the magnitude of this difference? This study seeks to answer these questions by conducting a comparative examination of crash incidence in lower-income and higher-income areas in Orange County, Florida. We then examine the nature of these differences and discuss their implications for equitable transportation system design.

1.1. Considering crash risk in lower-income areas

The general manner in which the built environment influences crash outcomes seems to be well understood; yet, this does little to explain the unique risk encountered in lower-income communities. For instance, the most consistent finding is the hazard posed by urban arterials, with such streets being associated with an increased incidence of traffic-related crashes, injuries, and deaths (Alluri et al., 2017; Eluru et al., 2016; Hadayeghi et al., 2007; Ladrón de Guevara et al., 2004; Lovegrove and Sayed, 2006; Tasic and Porter, 2016; Wang et al., 2012; Wang et al., 2016). At the same time, a recent study of dangerous pedestrian hotspots in the US found that they overwhelmingly occur along arterial corridors and, of these, the majority are located in lower-income areas and minority areas (Schneider et al., 2021).

Sidewalks, while often presumed to be a safety feature, have been found to have a mixed effect on crash incidence. Studies of pedestrian crashes reveal that the presence of sidewalks is often positively associated with pedestrian-related crashes (Cai et al., 2017; Eluru et al., 2016; Nashad et al., 2016). This counterintuitive finding is typically explained in terms of exposure; areas with more sidewalks are presumed to have more pedestrians, and thus more opportunities for pedestrian-involved crashes. Yet, a recent review of the literature found that few studies have meaningfully distinguished between pedestrian risk and pedestrian exposure (Merlin et al., 2020), rendering such explanations little more than professional conjecture.

Intersections, which are locations where opposing streams of traffic meet, are often associated with increased crash incidence (Abdel-Aty et al., 2013; Abdel-Aty et al., 2011; Dumbaugh and Rae, 2009; Dumbaugh et al., 2020; Hadayeghi et al., 2007). This is typically attributed to the fact that intersections are locations of conflict points between different streams of movement. Nonetheless, a pair of studies by Marshall and Garrick (2010; 2011) suggest that the relationship is somewhat more complex. In older communities, with dense networks comprised of two-way streets and smaller arterials, the presence of intersections was found to be associated with significantly fewer severe injury and fatal crashes than their more conventionally-designed counterparts.

Land use appears to matter as well, with the presence of commercial uses, such as retail establishments, being associated with higher rates of traffic-related crashes, deaths, and injuries (Dumbaugh and Rae, 2009; Dumbaugh and Li, 2010; Dumbaugh et al., 2020; Hadayeghi et al., 2007; Jermprapai and Srinivasan, 2014; Mohamed et al., 2014; Ukkusuri et al., 2011; Wier et al., 2009). This is unsurprising because commercial uses are major trip attractors that lead to higher traffic volumes and increased walking. The result is increased opportunities for traffic conflicts between road users, particularly at intersecting driveways.

This study investigates these issues via negative binomial regression models for lower-income and higher-income block groups in Orange County, Florida.

2. Material and methods

This study stratified Census block groups in Orange County, Florida, by median household income and aggregated crashes to that geography using the near analysis function in ArcGIS. Income criteria based on household size, established by the Department of Housing and Urban Development, were used to categorize block groups as being lower- or higher-income. For Orange County, the area median household income (AMI) was \$51,586 in 2016. Households making greater than 120 % of the AMI are generally ineligible for public assistance, leading us to categorize higher-income block groups as being those with median incomes of \$65,000 or greater. Median household incomes of \$40,000 or less were then identified as the threshold for lower-income block groups. This income framework divides the county approximately into thirds. Of the 374 total block groups in Orange County, 105 have median incomes of \$65,000 and greater, and 115 have median incomes of \$40,000 or less.

2.1. Built environment and socio-economic status data

To obtain information on community characteristics, this study uses data from the American Community Survey for the 2013–2017 period, as well as information on street characteristics and land use obtained from the Florida Geographic Data Library. These were used to calculate the following variables.

Income (in thousands):

Income is the median household income of the census block group. This was converted to thousands of dollars to simplify the interpretation of the model coefficients.

Population (in thousands):

Areas with more people have more trips and thus more opportunities for crash events. Information on population at the block group levels comes from the American Community Survey and is converted to thousands for ease of interpretation.

Percent black population:

Race is often poorly-accounted for in traffic safety research. Black Americans are often disproportionately affected by health disparities, such as cancer, strokes, and heart disease, among numerous others. This study sought to understand whether, after controlling for differences in income, Black residents were similarly disadvantaged in terms of safety outcomes.

Average annual daily traffic (AADT):

AADT data were provided by the Florida Department of Transportation for state highways. This variable is the sum of the AADT for surface highway segments located within the block group boundaries. AADT data for limited access facilities (i.e., freeways and interstates) are excluded from these counts.

Miles of urban arterials:

To calculate the mileage of urban arterials, this study used GIS to identify the length of roadways classified as urban arterials within the block group boundaries, and then summed the length of these arterial segments.

Number of commercial parcels:

The number of commercial parcels is the sum of unique parcels identified as being of commercial use.

Number of intersections:

This variable is the total number of intersections within a block group, identified using the streets shapefile.

Percent of streets with a sidewalk:

This variable is the sum of the mileage of streets with a sidewalk on at least one side, divided by the total mileage of streets in the block group.

Percent of sidewalks with a buffer:

The Florida Department of Transportation identifies whether a buffer is present between the vehicle travelway and sidewalks and shared paths. According to FDOT's classification, a buffer is defined as being a parking lane, a planting strip with objects such as trees and utility poles spaced less than 60 feet apart, or a guardrail. Such buffers are commonly regarded as being a pedestrian amenity as they physically separate pedestrians using sidewalks from vehicles traveling along a street. This variable is calculated as the miles of sidewalks buffered from oncoming traffic divided by the total miles of sidewalk in the block group.

Table 1 presents the descriptive statistics for these variables.

2.2. Crash data

This study uses crash data from the Florida Department of Transportation for the 2014 – 2016 period. Because this study is focused on the role of the built environment on crash incidence, it examines crashes occurring on surface streets and excludes crashes occurring on limited access facilities such as interstates and freeways. Three sets of variables are used in this analysis. The first is an examination of the total crashes occurring in each block group. In order to understand whether injurious crashes have unique characteristics, this study further examines KAB crashes, defined as fatal (K), incapacitating injury crashes (A), and non-incapacitating injury crashes (B). Possible injury crashes (C) were excluded in our injury counts as they are prone to inaccuracy. For example, a person may report feeling pain to the officer at the scene, leading to a determination of possible injury, though this may be short-lived. Alternatively, soft-tissue injuries or concussions may take hours or days to develop following an impact and are thus not identified as an injury in the police report. Finally, as pedestrian crashes are commonly a concern for lower-income populations, crashes involving pedestrians were identified as a third crash type for analysis.

Table 1
Descriptive Statistics.

Variable	All Block Groups			High-income Block Groups			Low-income Block Groups		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
Population (thousands)	3.45	0.09	31.53	3.83	0.27	31.53	2.97	0.09	9.56
Income (thousands)	56.86	0.00	222.7	95.06	65.56	222.7	29.30	0.00	39.95
% Black Population	21.07 %	0.00 %	98.07 %	6.08 %	0.00 %	35.43 %	42.50 %	0.00 %	98.07 %
AADT (thousands) (veh/day)	19.64	0.00	58.75	2.4	58.75	16.75	4.99	42.98	21.10
Miles of Urban Arterials	2.30	0.00	25.52	2.40	0.00	25.52	1.83	0.00	7.47
# of Commercial Uses	33.52	0.00	1143	25.5	0.00	540	30.7	0.00	517
# of Intersections	107.23	20	2371	134.30	26	2371	84.97	21.00	581
% of Streets with a Sidewalk	39.24 %	0.00 %	100.00 %	34.02 %	0.00 %	100.00 %	45.51 %	0.00 %	100.00 %
% of Sidewalks with a Buffer	16.72 %	0.00 %	100.00 %	15.76 %	0.00 %	100.00 %	12.50 %	0.00 %	100.00 %

2.3. Statistical models

This study thus examines total, KAB, and pedestrian crashes at three levels. It begins by examining the relationship between the built environment and crash incidence for the county as a whole in order to determine whether the general findings for Orange County are similar to that of other regions, thereby allowing us to ascertain whether or not our other findings may be more broadly generalizable. It then proceeds to examine crash incidence in lower-income and higher-income block groups separately to understand whether there are differences between them, and how they relate to crash patterns observed for the region as a whole. The distribution of block groups by median household income is shown in Fig. 1.

Because the dependent variables are count data that are over-dispersed (i.e., the variance is greater than the mean) as shown in Table 2, negative binomial regression models were used for this analysis. Negative binomial models are the more generalized version of the Poisson model. They account for overdispersion via the addition of stochastic random error in the Poisson function (Long, 1997). Because of this, negative binomial models tend to be frequently used to analyze discrete and infrequent events such as crash counts (Marshall and Ferenchak, 2019).

We applied the negative binomial model to our total dataset first and then separately to the higher- and lower-income block groups. For each group, we built three statistical models. The dependent variable for the first set was total crashes, including all severities. The second set of models focused on KAB crashes. In other words, the dependent variable included only fatal or injury (both incapacitating and non-incapacitating) crashes. The third set of models concentrated on the total number of pedestrian-vehicle crashes. The results systematically step through these three models for each of the three groups.

3. Results

3.1. Crash incidence in Orange County

Overall, our findings for Orange County, Florida, are consistent with previous research. As shown in Table 3, income had a strong and negative relationship with total, KAB, and pedestrian crashes. Similarly, more people, more AADT, more miles of urban arterials, and more commercial parcels were all positively and significantly related to increases in all three crash types. These findings are identical to those observed of other regions (Dumbaugh and Rae, 2009; Dumbaugh and Li, 2010; Dumbaugh et al., 2020), suggesting that our findings for lower-income and higher-income populations may reflect broader safety trends, rather than unique local conditions.

3.2. Crash incidence in higher-income block groups

Table 4 shows the model results for areas with median incomes of greater than \$65,000. Income remains significantly related to total and KAB crashes but ceases to be significantly associated with pedestrian

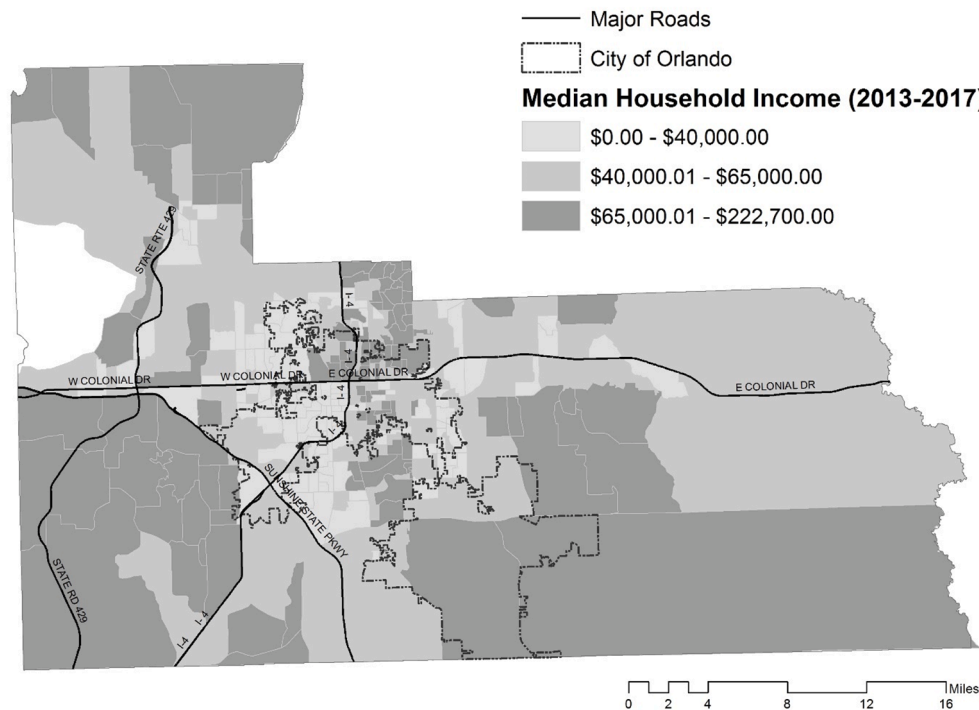


Fig. 1. Block Groups by Median Household Income, Orange County.

Table 2
Mean and Dispersion of Dependent Variables.

	Orange County		Higher-income BGs		Lower-income BGs	
	Mean	Std. Dev.	Mean	Std. Dev.	Mean	Std. Dev.
Total Crashes	195	227	143	191	196	182
KAB Crashes	38	37	30	41	39	33
Pedestrian Crashes	4	4	2	3	5	5

Table 3
Total, KAB, and Pedestrian Crash Models for All Block Groups.

Orange County All Block Groups	Total		KAB		Pedestrian	
	Coeff.	Sig.	Coeff.	Sig.	Coeff.	Sig.
Population (Thousands)	0.049	0.000	0.070	0.000	0.029	0.000
Income (Thousands)	-0.008	0.000	-0.008	0.000	-0.011	0.000
% Black Population	0.002	0.183	0.003	0.100	0.006	0.100
AADT (Thousands)	0.022	0.000	0.028	0.000	0.023	0.000
Miles of Urban Arterials	0.166	0.000	0.179	0.000	0.061	0.000
# of Commercial Uses	0.002	0.007	0.001	0.013	0.002	0.013
# of Intersections	0.001	0.136	0.000	0.394	0.001	0.394
% of Streets with a Sidewalk	0.001	0.723	-0.001	0.361	0.000	0.361
% of Sidewalks with a Buffer	0.436	0.002	0.245	0.071	0.320	0.071
Constant	4.150	0.000	2.442	0.000	0.718	0.000
n = 374	Log Likelihood = -2188		Log Likelihood = -1564		Log Likelihood = -841	

Table 4
Total, KAB, and Pedestrian Crash Models for Block Groups with Median Household Incomes of \$65,000 and Greater.

Orange County \$65,000 and Greater	Total		KAB		Pedestrian	
	Coeff.	P(z)	Coeff.	P(z)	Coeff.	P(z)
Population (Thousands)	0.075	0.000	0.074	0.000	0.062	0.006
Income (Thousands)	-0.008	0.001	-0.006	0.008	-0.005	0.280
% Black Population	0.002	0.867	0.015	0.215	-0.007	0.691
AADT (Thousands)	0.014	0.197	0.020	0.055	0.006	0.672
Miles of Urban Arterials	0.083	0.012	0.087	0.004	-0.019	0.632
# of Commercial Uses	0.003	0.045	0.002	0.070	0.004	0.016
# of Intersections	0.000	0.964	0.000	0.605	0.000	0.430
% of Streets with a Sidewalk	0.003	0.397	0.003	0.253	0.001	0.805
% of Sidewalks with a Buffer	0.798	0.013	0.635	0.035	0.815	0.055
Constant	4.314	0.000	2.269	0.000	0.266	0.616
n = 105	Log Likelihood = -572		Log Likelihood = -397		Log Likelihood = -175	

crashes. AADT proved to be significantly associated only with KAB crashes, but not total or pedestrian crashes. The characteristics of the built environment prove more useful in understanding differences in crash incidence. The number of commercial uses in a block group was associated with statistically-significant increases in total, KAB, and pedestrian crashes. The mileage of urban arterials in a block group was likewise associated with a significant increase in total and KAB crashes. Arterials, while significantly-related to increases in total and KAB crashes, nevertheless have a weaker relationship with crash incidence than they do for the region as a whole.

The mileage of arterials was not associated with pedestrian crashes in affluent areas, though the presence of buffered sidewalks were. These unexpected findings are likely attributable to exposure; that is,

affluent households likely have access to a personal automobile, rendering decisions to walk a matter of choice, rather than necessity. Indeed, the findings for pedestrian crashes as a whole suggest that features associated with higher rates of walking – more people, more nearby commercial destinations, and buffered sidewalks – are also associated with increased pedestrian crashes. The latter finding is particularly notable since buffered sidewalks are generally regarded as a pedestrian safety feature. Yet in the case of Orlando, a 1 % increase in the number of streets with a sidewalk buffer not only corresponded to an 82 % increase in pedestrian crashes, it was also associated with an 80 % increase in total crashes and a 64 % increase in KAB crashes.

3.3. Crash incidence in lower-income block groups

Table 5 presents the model results for lower-income block groups, defined as block groups with median household incomes of \$40,000 and less. Median household income and the number of intersections proved to be comparatively unimportant as explanatory variables, while the number of people residing in a block group is associated with significant increases in total and KAB crashes.

Nonetheless, the remaining variables differ in notable ways from those of higher-income block groups. One of the most notable pertains to race; while the percentage of Black residents of a block group did not have statistically-meaningful relationships with any of the crash types in higher income block groups, this variable was associated with significantly higher rates of total, KAB, and pedestrian crashes alike. Stated another way, areas with higher concentrations of Black populations are at increased risk of being involved in a crash, even after accounting for any increased risk associated with differences in income.

Unlike more affluent areas, where background traffic volumes were not meaningfully related to pedestrian crashes, AADT was significantly associated with increases in total, KAB, and pedestrian crashes in lower-income areas. The miles of urban arterials proved particularly hazardous in lower-income communities. Not only were the mileage of urban arterials significantly associated with increases in all three crash types, they also reported coefficients for total and KAB crashes that are roughly-three times greater than those of higher-income populations. And unlike in affluent areas, where arterials had no statistically meaningful effect on pedestrian crashes, in lower-income areas, each mile of urban arterial was associated with a 15 % increase in pedestrian crashes.

While the presence of sidewalk buffers proved to be associated with strong increases in the incidence of total, injurious, and pedestrian crashes for higher-income block groups, they were generally associated

with reductions in injurious and pedestrian crashes in lower-income areas, though not at conventional levels of statistical significance. Finally, while the number of commercial uses was significantly related to all three crash types in affluent communities, they were only significantly related to increases in total crashes for lower-income areas, though it must be acknowledged that these variables are very close to the 0.10 significance level.

4. Discussion

While previous research has consistently identified income as a significant factor in understanding community-level crash risk, the results of this study suggest that income has a far more complex relationship to crash incidence than has been previously supposed, with environmental features affecting lower-income and higher-income areas in notably different ways. Table 6, below, compares the environmental risk factors for lower-income and affluent areas, followed by a discussion of our major findings.

First, while urban arterials are a risk factor for all areas, their negative safety effect is profoundly greater in lower-income environments. Each mile of urban arterial in a lower-income community is associated with three times the number of total, injurious, and fatal crashes as they are in more affluent areas. Moreover, while arterials had a negative, but statistically-insignificant effect on pedestrian crashes in affluent areas, they were associated in significant increases in pedestrian crashes in lower-income ones.

At least superficially, there appears to be little difference in the overall design of arterials traversing these environments; access management, which entails the use of raised medians to remove conflicts associated with commercial driveways, is consistently applied along arterial in lower-income and more affluent areas alike. An examination of the geography of lower-income communities provides a likely explanation (see Fig. 1). Lower-income areas tend to congregate immediately outside the urban core. As such, it is very likely that a substantial share of the arterial traffic in lower-income areas consists of persons who reside elsewhere and are travelling through these areas to access downtown Orlando, persons who may exhibit less overall caution than they might in their own neighborhoods. If so, then regional development patterns, which allow higher-income neighborhoods to be established on the periphery of urban areas, may be a contributing factor to crash incidence in lower-income areas. The findings for AADT support this interpretation; lower-income areas, while having slightly fewer total miles of arterials than the higher-income communities, nevertheless report fully twice the total AADT (see Table 1), strongly suggesting that much of traffic in these areas consists of persons who reside elsewhere. Nonetheless, our data do not allow us to ascertain the residential locations of persons involved in the crashes that occur in these areas; future research will be needed to confirm this hypothesis.

A second notable difference is the manner in which street network characteristics affect pedestrian crashes in these areas. For lower-income areas, each additional mile of arterial thoroughfare is associated with a 15 % increase in pedestrian crashes, a not unexpected finding given the overall hazard associated with this road class. Yet for affluent areas, each additional mile of arterial thoroughfare is associated with a reduction in pedestrian crashes, though not at statistically meaningful levels. Also surprising is the finding that the presence of buffers between the sidewalk and the vehicle travelway, which are generally viewed as pedestrian safety features, have markedly different effects on pedestrian crashes in lower-income and affluent areas. For lower income areas, sidewalk buffers are associated with a reduction in pedestrian crashes, though not at statistically meaningful levels. In affluent areas, however pedestrians are associated with a significant increase in pedestrian crashes. To state the problem succinctly, in affluent areas, features thought to be hazardous for pedestrians—arterials—proved not to be, while features presumed to enhance the safety of pedestrians—sidewalk buffers—proved to be associated with an increase in pedestrian crashes.

Table 5
Total, KAB, and Pedestrian Crash Models for Block Groups with Median Household Incomes of \$40,000 and Less.

Orange County \$40,000 and Less	Total		KAB		Pedestrian	
	Coeff.	P(z)	Coeff.	P(z)	Coeff.	P(z)
Population (Thousands)	0.079	0.018	0.099	0.003	0.078	0.003
Income (Thousands)	−0.007	0.297	−0.003	0.697	−0.004	0.697
% Black Population	0.003	0.054	0.004	0.018	0.010	0.018
AADT (Thousands)	0.015	0.065	0.024	0.004	0.027	0.004
Miles of Urban Arterials	0.265	0.000	0.244	0.000	0.154	0.000
# of Commercial Uses	0.002	0.074	0.002	0.111	0.001	0.111
# of Intersections	0.001	0.383	0.001	0.540	0.001	0.540
% of Streets with a Sidewalk	0.000	0.845	−0.003	0.155	−0.004	0.155
% of Sidewalks with a Buffer	0.280	0.287	−0.276	0.305	−0.251	0.305
Constant	3.915	0.000	2.153	0.000	0.185	0.000
n = 115	Log Likelihood = −661		Log Likelihood = −483		Log Likelihood = −276	

Table 6

A Comparison of Risk Factors for High-Income and Low-Income Block Groups.

Variable	High Income			Low Income		
	Total	KAB	Pedestrian	Total	KAB	Pedestrian
Population (Thousands)	0.075***	0.074***	0.062**	0.079*	0.099**	0.078**
Income (Thousands)	-0.008***	-0.006**	-0.005	-0.007	-0.003	-0.004
% Black Population	0.002	0.015	-0.007	0.003^ψ	0.004**	0.010**
AADT (Thousands)	0.014	0.020^ψ	0.006	0.015^ψ	0.024**	0.027**
Miles of Urban Arterials	0.083**	0.087*	-0.019	0.265***	0.244***	0.154***
# of Commercial Uses	0.003*	0.002^ψ	0.004*	0.002^ψ	0.002	0.001
# of Intersections	0.000	0.000	0.000	0.001	0.001	0.001
% of Streets with a Sidewalk	0.003	0.003	0.001	0.000	-0.003	-0.004
% of Sidewalks with a Buffer	0.798*	0.635*	0.815^ψ	0.280	-0.276	-0.251

*p ≤ 0.05.

**p ≤ 0.01.

***p ≤ 0.001.

^ψp ≤ 0.1.

These differences begin to make sense once one considers differences in the purposes for which lower-income and more affluent households undertake walking trips. Studies examining these differences found that walking for lower-income households is typically undertaken for utilitarian purposes, such as travelling to work or school. These are necessary trips that will be undertaken regardless of the quality of the built environment (Kang et al., 2017; Yang and Diez Roux, 2012) and, as a result, when they occur in areas with hazardous thoroughfares, such as arterials, would be expected to result in an increase in pedestrian crashes. Persons in affluent households, by contrast, undertake walk trips principally for recreational purposes such as exercise or dog walking (Yang and Diez Roux, 2012). Unpleasant or unsafe environments, such as arterials, can be readily avoided by choosing routes that avoid these environments, driving to another, more accommodating environment to undertake the walk trip, or foregoing the walk trip altogether.

We strongly suspect that these differences in the purposes of walking trips explain the differences in our findings for pedestrian buffers as well. Locations with sidewalks that are buffered from oncoming traffic feel safer than those without, which in turn results in an increase in pedestrian exposure due to an increase in recreational walking. Unfortunately, our data do not provide details on the characteristics of these buffers, nor do we have information on pedestrian volumes, the combination of which prevents us from further exploring this finding.

Perhaps the most noteworthy finding of this study is that race—and the underlying cultural conditions that it implies—plays a far more prominent role in understanding crash incidence than is generally assumed. While the percentage of Black residents in a block group had no meaningful relationship to crash incidence in more affluent areas, it was significantly related to increases in total, KAB, and pedestrian crashes. This relationship was particularly notable with pedestrian crashes, with a 1 % increase in the percentage of Black residents in a lower-income community being associated with a 1 % increase in pedestrian crashes. This is compounded by the concentration of Black residents in lower-income communities; while 21 % of Orange County's residents identify as Black, Black persons are disproportionately located in lower income communities. The current study does not allow us to explain this finding, though we suspect that a combination of two factors are likely. The first is that elected officials and transportation professionals are failing to pay adequate attention to the pedestrian safety needs of predominantly Black communities, resulting in safety problems being unaddressed. The second is that cultural or societal factors may result in behaviors that increase crash risk for black populations. A study examining driver yielding to pedestrians, for example, found that motorists were twice as likely to yield for white pedestrian than black ones (Goddard et al., 2015). As such, racial bias may translate into increased crash risk in areas with concentrations of non-white populations. Future research is needed to better understand the safety disparity confronting

Black communities.

5. Conclusions

To date, there has been a general tendency to regard the built environment as having consistent safety impacts on all races and incomes alike. Yet, as detailed here, this cannot be presumed to be the case. Traffic safety and the design of the built environment, like much of U.S. society, appears to be strongly intertwined with broader issues of racial and income inequality. Arterials are associated with three times as many deaths and injuries in lower-income environments than in more affluent ones. The difference in these outcomes appears to be less a matter of design, and more the result of lower-income populations being located along major regional arteries, which carry far higher traffic volumes than the more affluent areas located on the periphery of the County. Moreover, while arterials are associated with increased pedestrian crashes in lower-income environments, they did not prove to be meaningfully associated with crashes in more affluent ones. By contrast, the presence of sidewalk buffers, normally regarded as a pedestrian safety feature, were associated with increases in pedestrian crashes in affluent areas, though not for lower-income ones. We expect that these divergent findings are attributable to income-related differences in pedestrian activity. For lower-income areas, walking is undertaken principally for utilitarian purposes, while in affluent ones, walking is a recreational activity. As such, affluent pedestrians can avoid routes viewed as hazardous, such as arterial highways, and divert to routes that are more accommodating, such as those with sidewalk buffers. Finally, and notably, lower-income areas with higher concentration of Black residents were associated with increases in traffic-related deaths and injuries, even after accounting for the effects of income, indicating a racial disparity that is currently unaddressed in either research or practice. These results identify a clear need for road safety professionals to better account for income inequality and race when designing for better traffic safety outcomes.

We conclude by noting some of the limitations of this study. First and foremost, these data are aggregate in nature. The focus on block group-level safety outcomes may mask localized differences in design. For example, the data do not provide information on the quality of the sidewalks present in these communities or whether the sidewalk buffers contain fixed objects such as street trees or parked cars, which may provide a physical separation between the sidewalk and the vehicle travelway. It may well be the case that different combinations of sidewalks and sidewalk buffers influence safety differently, particularly for pedestrian crashes. It must also be recognized that this study has focused on differences in crash outcomes and has sought to infer the reasons for the differences. There is a need to formally examine not only difference in crash outcomes, but also differences in the underlying conditions that may explain them. Nonetheless, this study's findings highlight the need

to better address the manner in which social vulnerabilities place lower income and especially Black populations at disproportionate risk.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Eric Dumbaugh reports financial support was provided by United States Department of Transportation Research and Innovative Technology Administration.

Data availability

Data will be made available on request.

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